MILITARY SPECIFICATION SHEET

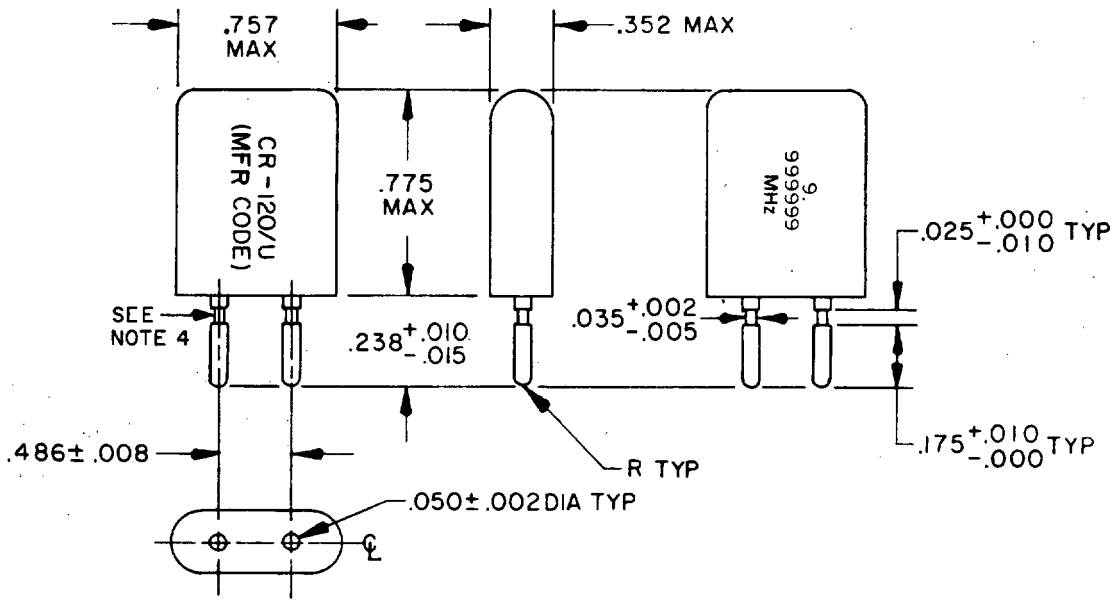
CRYSTAL UNIT, QUARTZ, CR-120/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(D) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 2 to 10 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type designation, frequency, and code designation characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. The pin undercut may be omitted.

Figure 1. Crystal unit - CR-120/U.

(D) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-27/U (MIL-H-10056/14)
Dimensions, marking, and configuration	See figure 1
Frequency range	2 to 10 MHz, inclusive
Frequency tolerance	±5 ppm at 85°C ± 5°C
Antiresonance load capacitance	100 ±0.5 pF
Mode of oscillation	Fundamental
Temperature range:	
Operating	85°C ± 5°C
Operable	-55°C to +90°C
Test set	TS-330/TSM, modified with RFL AGC adapter, model HB-16940 for low drive application
Drive level, rated	2 µW
Crystal quality (Q) 1/	2 to 5 MHz, 1 x 10^6 minimum 5+ to 10 MHz, .5 x 10^6 minimum
Frequency temperature coefficient:	
Turning point in the range of	85°C ±5°C
Turning point to be marked on holder	To the nearest 0.1°C
Power-frequency coefficient	±0.01 ppm maximum from 1 to 3µW
Shock (specified pulse):	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Vibration, high frequency: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent
Low temperature storage:	
Resistance change permitted	±15 percent
Aging: 3/	2 to 5 MHz, 0.01 ppm per week for the last 2 weeks of a 4-week aging test. 5+ to 10 MHz, 0.02 ppm per week for the last 2 weeks of a 4-week aging test.

1/ $Q = \frac{1}{4\pi (\Delta f)(C_T)(R)}$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 100 pF.

C_T = Capacitance: C_0 (shunt capacitance of the crystal unit) plus C_L (load capacitance) of 100 pF.

R = Motional-arm resistance of the crystal unit.

2/ Method 204 of MIL-STD-202.

3/ The crystal units shall be preconditioned by storage in an oven at 85°C ±5°C for 3 weeks at the time of manufacture.

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Ⓓ QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE:

NATO guidance type NXT 46.

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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